

the fact that $\Gamma(1) = 1$ yields

$$\begin{aligned}
 e^{-A} &= \prod_{n=1}^{\infty} \left(1 + \frac{1}{n}\right) e^{-1/n} \\
 &= \lim_{N \rightarrow \infty} \prod_{n=1}^N \left(1 + \frac{1}{n}\right) e^{-1/n} \\
 &= \lim_{N \rightarrow \infty} e^{\sum_{n=1}^N [\log(1+1/n) - 1/n]} \\
 &= \lim_{N \rightarrow \infty} e^{-(\sum_{n=1}^N 1/n) + \log N + \log(1+1/N)} \\
 &= e^{-\gamma}.
 \end{aligned}$$

Therefore $A = \gamma + 2\pi ik$ for some integer k . Since $\Gamma(s)$ is real whenever s is real, we must have $k = 0$, and the argument is complete.

Note that the proof shows that the function $1/\Gamma$ is essentially characterized (up to two normalizing constants) as the entire function that has:

- (i) simple zeros at $s = 0, -1, -2, \dots$ and vanishes nowhere else, and
- (ii) order of growth ≤ 1 .

Observe that $\sin \pi s$ has a similar characterization (except the zeros are now at *all* the integers). However, while $\sin \pi s$ has a stricter growth estimate of the form $\sin \pi s = O(e^{c|s|})$, this estimate (without the logarithm in the exponent) does not hold for $1/\Gamma(s)$ as Exercise 12 demonstrates.

2 The zeta function

The Riemann **zeta function** is initially defined for real $s > 1$ by the convergent series

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

As in the case of the gamma function, ζ can be continued into the complex plane. There are several proofs of this fact, and we present in the next section the one that relies on the functional equation of ζ .

2.1 Functional equation and analytic continuation

In parallel to the gamma function, we first provide a simple extension of ζ to a half-plane in $\mathbb{C}$.

Proposition 2.1 *The series defining $\zeta(s)$ converges for $\operatorname{Re}(s) > 1$, and the function ζ is holomorphic in this half-plane.*

Proof. If $s = \sigma + it$ where σ and t are real, then

$$|n^{-s}| = |e^{-s \log n}| = e^{-\sigma \log n} = n^{-\sigma}.$$

As a consequence, if $\sigma > 1 + \delta > 1$ the series defining ζ is uniformly bounded by $\sum_{n=1}^{\infty} 1/n^{1+\delta}$, which converges. Therefore, the series $\sum 1/n^s$ converges uniformly on every half-plane $\operatorname{Re}(s) > 1 + \delta > 1$, and therefore defines a holomorphic function in $\operatorname{Re}(s) > 1$.

The analytic continuation of ζ to a meromorphic function in $\mathbb{C}$ is more subtle than in the case of the gamma function. The proof we present here relates ζ to Γ and another important function.

Consider the **theta function**, already introduced in Chapter 4, which is defined for real $t > 0$ by

$$\vartheta(t) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t}.$$

An application of the Poisson summation formula (Theorem 2.4 in Chapter 4) gave the functional equation satisfied by ϑ , namely

$$\vartheta(t) = t^{-1/2} \vartheta(1/t).$$

The growth and decay of ϑ we shall need are

$$\vartheta(t) \leq C t^{-1/2} \quad \text{as } t \rightarrow 0,$$

and

$$|\vartheta(t) - 1| \leq C e^{-\pi t} \quad \text{for some } C > 0, \text{ and all } t \geq 1.$$

The inequality for t tending to zero follows from the functional equation, while the behavior as t tends to infinity follows from the fact that

$$\sum_{n \geq 1} e^{-\pi n^2 t} \leq \sum_{n \geq 1} e^{-\pi n t} \leq C e^{-\pi t}$$

for $t \geq 1$.

We are now in a position to prove an important relation among ζ , Γ and ϑ .

Theorem 2.2 *If $\operatorname{Re}(s) > 1$, then*

$$\pi^{-s/2}\Gamma(s/2)\zeta(s) = \frac{1}{2} \int_0^\infty u^{(s/2)-1} [\vartheta(u) - 1] du.$$

Proof. This and further arguments are based on the observation that

$$(6) \quad \int_0^\infty e^{-\pi n^2 u} u^{(s/2)-1} du = \pi^{-s/2}\Gamma(s/2)n^{-s}, \quad \text{if } n \geq 1.$$

Indeed, if we make the change of variables $u = t/\pi n^2$ in the integral, the left-hand side becomes

$$\left(\int_0^\infty e^{-t} t^{(s/2)-1} dt \right) (\pi n^2)^{-s/2},$$

which is precisely $\pi^{-s/2}\Gamma(s/2)n^{-s}$. Next, note that

$$\frac{\vartheta(u) - 1}{2} = \sum_{n=1}^\infty e^{-\pi n^2 u}.$$

The estimates for ϑ given before the statement of the theorem justify an interchange of the infinite sum with the integral, and thus

$$\begin{aligned} \frac{1}{2} \int_0^\infty u^{(s/2)-1} [\vartheta(u) - 1] du &= \sum_{n=1}^\infty \int_0^\infty u^{(s/2)-1} e^{-\pi n^2 u} du \\ &= \pi^{-s/2}\Gamma(s/2) \sum_{n=1}^\infty n^{-s} \\ &= \pi^{-s/2}\Gamma(s/2)\zeta(s), \end{aligned}$$

as was to be shown.

In view of this, we now consider the modification of the ζ function called the **xi function**, which makes the former appear more symmetric. It is defined for $\operatorname{Re}(s) > 1$ by

$$(7) \quad \xi(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s).$$

Theorem 2.3 *The function ξ is holomorphic for $\operatorname{Re}(s) > 1$ and has an analytic continuation to all of $\mathbb{C}$ as a meromorphic function with simple poles at $s = 0$ and $s = 1$. Moreover,*

$$\xi(s) = \xi(1-s) \quad \text{for all } s \in \mathbb{C}.$$

Proof. The idea of the proof is to use the functional equation for ϑ , namely

$$\sum_{n=-\infty}^{\infty} e^{-\pi n^2 u} = u^{-1/2} \sum_{n=-\infty}^{\infty} e^{-\pi n^2 / u}, \quad u > 0.$$

We then could multiply both sides by $u^{(s/2)-1}$ and try to integrate in u . Disregarding the terms corresponding to $n = 0$ (which produce infinities in both sums), we would get the desired equality once we invoked formula (6), and the parallel formula obtained by making the change of variables $u \mapsto 1/u$. The actual proof requires a little more work and goes as follows.

Let $\psi(u) = [\vartheta(u) - 1]/2$. The functional equation for the theta function, namely $\vartheta(u) = u^{-1/2}\vartheta(1/u)$, implies

$$\psi(u) = u^{-1/2}\psi(1/u) + \frac{1}{2u^{1/2}} - \frac{1}{2}.$$

Now, by Theorem 2.2 for $\operatorname{Re}(s) > 1$, we have

$$\begin{aligned} \pi^{-s/2}\Gamma(s/2)\zeta(s) &= \int_0^{\infty} u^{(s/2)-1}\psi(u) du \\ &= \int_0^1 u^{(s/2)-1}\psi(u) du + \int_1^{\infty} u^{(s/2)-1}\psi(u) du \\ &= \int_0^1 u^{(s/2)-1} \left[u^{-1/2}\psi(1/u) + \frac{1}{2u^{1/2}} - \frac{1}{2} \right] du + \\ &\quad + \int_1^{\infty} u^{(s/2)-1}\psi(u) du \\ &= \frac{1}{s-1} - \frac{1}{s} + \int_1^{\infty} (u^{(-s/2)-1/2} + u^{(s/2)-1}) \psi(u) du \end{aligned}$$

whenever $\operatorname{Re}(s) > 1$. Therefore

$$\xi(s) = \frac{1}{s-1} - \frac{1}{s} + \int_1^{\infty} (u^{(-s/2)-1/2} + u^{(s/2)-1}) \psi(u) du.$$

Since the function ψ has exponential decay at infinity, the integral above defines an entire function, and we conclude that ξ has an analytic continuation to all of $\mathbb{C}$ with simple poles at $s = 0$ and $s = 1$. Moreover, it is

immediate that the integral remains unchanged if we replace s by $1 - s$, and the same is true for the sum of the two terms $1/(s - 1) - 1/s$. We conclude that $\xi(s) = \xi(1 - s)$ as was to be shown.

From the identity we have proved for ξ we obtain the desired result for the zeta function: its analytic continuation and its functional equation.

Theorem 2.4 *The zeta function has a meromorphic continuation into the entire complex plane, whose only singularity is a simple pole at $s = 1$.*

Proof. A look at (7) provides the meromorphic continuation of ζ , namely

$$\zeta(s) = \pi^{s/2} \frac{\xi(s)}{\Gamma(s/2)}.$$

Recall that $1/\Gamma(s/2)$ is entire with simple zeros at $0, -2, -4, \dots$, so the simple pole of $\xi(s)$ at the origin is cancelled by the corresponding zero of $1/\Gamma(s/2)$. As a consequence, the only singularity of ζ is a simple pole at $s = 1$.

We shall now present a more elementary approach to the analytic continuation of the zeta function, which easily leads to its extension in the half-plane $\operatorname{Re}(s) > 0$. This method will be useful in studying the growth properties of ζ near the line $\operatorname{Re}(s) = 1$ (which will be needed in the next chapter). The idea behind it is to compare the sum $\sum_{n=1}^{\infty} n^{-s}$ with the integral $\int_1^{\infty} x^{-s} dx$.

Proposition 2.5 *There is a sequence of entire functions $\{\delta_n(s)\}_{n=1}^{\infty}$ that satisfy the estimate $|\delta_n(s)| \leq |s|/n^{\sigma+1}$, where $s = \sigma + it$, and such that*

$$(8) \quad \sum_{1 \leq n < N} \frac{1}{n^s} - \int_1^N \frac{dx}{x^s} = \sum_{1 \leq n < N} \delta_n(s),$$

whenever N is an integer > 1 .

This proposition has the following consequence.

Corollary 2.6 *For $\operatorname{Re}(s) > 0$ we have*

$$\zeta(s) - \frac{1}{s-1} = H(s),$$

where $H(s) = \sum_{n=1}^{\infty} \delta_n(s)$ is holomorphic in the half-plane $\operatorname{Re}(s) > 0$.

To prove the proposition we compare $\sum_{1 \leq n < N} n^{-s}$ with $\sum_{1 \leq n < N} \int_n^{n+1} x^{-s} dx$, and set

$$(9) \quad \delta_n(s) = \int_n^{n+1} \left[\frac{1}{n^s} - \frac{1}{x^s} \right] dx.$$

The mean-value theorem applied to $f(x) = x^{-s}$ yields

$$\left| \frac{1}{n^s} - \frac{1}{x^s} \right| \leq \frac{|s|}{n^{\sigma+1}}, \quad \text{whenever } n \leq x \leq n+1.$$

Therefore $|\delta_n(s)| \leq |s|/n^{\sigma+1}$, and since

$$\int_1^N \frac{dx}{x^s} = \sum_{1 \leq n < N} \int_n^{n+1} \frac{dx}{x^s},$$

the proposition is proved.

Turning to the corollary, we assume first that $\operatorname{Re}(s) > 1$. We let N tend to infinity in formula (8) of the proposition, and observe that by the estimate $|\delta_n(s)| \leq |s|/n^{\sigma+1}$ we have the uniform convergence of the series $\sum \delta_n(s)$ (in any half-plane $\operatorname{Re}(s) \geq \delta$ when $\delta > 0$). Since $\operatorname{Re}(s) > 1$, the series $\sum n^{-s}$ converges to $\zeta(s)$, and this proves the assertion when $\operatorname{Re}(s) > 1$. The uniform convergence also shows that $\sum \delta_n(s)$ is holomorphic when $\operatorname{Re}(s) > 0$, and thus shows that $\zeta(s)$ is extendable to that half-plane, and that the identity continues to hold there.

Remark. The idea described above can be developed step by step to yield the continuation of ζ into the entire complex plane, as shown in Problems 2 and 3. Another argument giving the full analytic continuation of ζ is outlined in Exercises 15 and 16.

As an application of the proposition we can show that the growth of $\zeta(s)$ near the line $\operatorname{Re}(s) = 1$ is “mild.” Recall that when $\operatorname{Re}(s) > 1$, we have $|\zeta(s)| \leq \sum_{n=1}^{\infty} n^{-\sigma}$, and so $\zeta(s)$ is bounded in any half-plane $\operatorname{Re}(s) \geq 1 + \delta$, with $\delta > 0$. We shall see that on the line $\operatorname{Re}(s) = 1$, $|\zeta(s)|$ is majorized by $|t|^\epsilon$, for every $\epsilon > 0$, and that the growth near the line is not much worse. The estimates below are not optimal. In fact, they are rather crude but suffice for what is needed later on.

Proposition 2.7 *Suppose $s = \sigma + it$ with $\sigma, t \in \mathbb{R}$. Then for each σ_0 , $0 \leq \sigma_0 \leq 1$, and every $\epsilon > 0$, there exists a constant c_ϵ so that*

$$(i) \quad |\zeta(s)| \leq c_\epsilon |t|^{1-\sigma_0+\epsilon}, \text{ if } \sigma_0 \leq \sigma \text{ and } |t| \geq 1.$$

(ii) $|\zeta'(s)| \leq c_\epsilon |t|^\epsilon$, if $1 \leq \sigma$ and $|t| \geq 1$.

In particular, the proposition implies that $\zeta(1+it) = O(|t|^\epsilon)$ as $|t|$ tends to infinity,³ and the same estimate also holds for ζ' . For the proof, we use Corollary 2.6. Recall the estimate $|\delta_n(s)| \leq |s|/n^{\sigma+1}$. We also have the estimate $|\delta_n(s)| \leq 2/n^\sigma$, which follows from the expression for $\delta_n(s)$ given by (9) and the fact that $|n^{-s}| = n^{-\sigma}$ and $|x^{-s}| \leq n^{-\sigma}$ if $x \geq n$. We then combine these two estimates for $|\delta_n(s)|$ via the observation that $A = A^\delta A^{1-\delta}$, to obtain the bound

$$|\delta_n(s)| \leq \left(\frac{|s|}{n^{\sigma_0+1}} \right)^\delta \left(\frac{2}{n^{\sigma_0}} \right)^{1-\delta} \leq \frac{2|s|^\delta}{n^{\sigma_0+\delta}},$$

as long as $\delta \geq 0$. Now choose $\delta = 1 - \sigma_0 + \epsilon$ and apply the identity in Corollary 2.6. Then, with $\sigma = \operatorname{Re}(s) \geq \sigma_0$, we find

$$|\zeta(s)| \leq \left| \frac{1}{s-1} \right| + 2|s|^{1-\sigma_0+\epsilon} \sum_{n=1}^{\infty} \frac{1}{n^{1+\epsilon}},$$

and conclusion (i) is proved. The second conclusion is actually a consequence of the first by a slight modification of Exercise 8 in Chapter 2. For completeness we sketch the argument. By the Cauchy integral formula,

$$\zeta'(s) = \frac{1}{2\pi r} \int_0^{2\pi} \zeta(s + re^{i\theta}) e^{i\theta} d\theta,$$

where the integration is taken over a circle of radius r centered at the point s . Now choose $r = \epsilon$ and observe that this circle lies in the half-plane $\operatorname{Re}(s) \geq 1 - \epsilon$, and so (ii) follows as a consequence of (i) on replacing 2ϵ by ϵ .

3 Exercises

1. Prove that

$$\Gamma(s) = \lim_{n \rightarrow \infty} \frac{n^s n!}{s(s+1) \cdots (s+n)}$$

whenever $s \neq 0, -1, -2, \dots$

[Hint: Use the product formula for $1/\Gamma$, and the definition of the Euler constant γ .]

2. Prove that

$$\prod_{n=1}^{\infty} \frac{n(n+a+b)}{(n+a)(n+b)} = \frac{\Gamma(a+1)\Gamma(b+1)}{\Gamma(a+b+1)}$$

³The reader should recall the O notation which was introduced at the end of Chapter 1.